



2 Frequency Geodesic Dome Calculator

2v Icosahedron

How to Use This Calculator

Select Hub Size

No Hubs: ☒ Standard 4" Hubs: ☐ Super 6" Hubs: ☐ Mega 8" Hubs: ☐

Feet: Meters: [Click to Calculate](#) Convert to Feet & Inches Chord

Enter the Number of Feet in **One Field**, then Click the "Calculate" Button.

9.7083	Dome Height (h)	9' 8 - 1/2"	
19.4165	Dome Width / Diameter (d)	19' 5"	
6	Length of 35 Blue "A" Struts	6' 0"	.61803
5.3059	Length of 30 Red "B" Struts	5' 3 - 5/8"	.54653
9.7083	Dome Radius (r)	9' 8 - 1/2"	
61	Dome Equator Circumference (2πr)	61' 0"	
296.1	Dome Circular Floor Area in Square Feet (πr²)		
592.19 ft²	Dome Spherical Surface Area in Square Feet (2πrh)		
1,916.39 ft³	Dome Spherical Volume in Cubic Feet (πh(3r² + h²)/6)		
10	Dome Base Perimeter Number of Sides		
60'	10 Sided Base Perimeter Length (Feet)		
276.99 ft²	10 Sided Base Floor Area (Square Feet)		
40	Total Number of Dome Panels / Faces		
549.75 ft²	Surface Area for 40 Dome Panels / Faces (Square Feet)		
1,673.86 ft³	Total Volume of Tetrahedrons behind 40 Dome Panels / Faces in Cubic Feet		
6	Number of Red 72° 5-Way Hubs		
20	Number of Blue 60° 6-Way Hubs		
26	Number of Center Rings		
130	Number of Zip Ties to Construct the Dome		
78	Number of Zip Ties to Secure the Inside Dual Covering Hubs (3 per hub)		
78	Number of Zip Ties to Secure the Outside Dual Covering Hubs (3 per hub)		
30.5' Length 30.5' Width	Absolute Minimum Square Covering Dimensions for a Tightly Fitted Dome Cover		
33.5' Length 33.5' Width	Minimum Square Covering Dimensions for a Dome Cover Secured to the Hubs with Zip Ties with a Minimum Overlap at the Bottom of the Dome		
36.6' Length 36.6' Width	Maximum Square Covering Dimensions for a Dome Cover Secured to the Hubs with Zip Ties with an Ample Overlap at the Bottom of the Dome		
369.18'	Total Length of All Struts Added Together (Feet)		

[Instructional Video on How to Use This Section of the Dome Calculator](#)

Cost Calculation and Material Utilization Chart

[Dome Calculator](#) [Cost of Materials](#) [Weight Chart](#) [Construction Diagram](#) [Dome Panels / Coverings](#)

• [Blue "A" Struts](#)
• [Red "B" Struts](#)
• [Grand Totals](#)

[Calculate](#) [Clear Fields](#)

Length of Pipe: Feet Cost Per Pipe:

Blue "A" Struts

Blue "A" Strut Units Entered: Feet and Inches: "

Number of Pipes needed to Create (35) Blue "A" Struts:

Number of Blue "A" Struts Produced From Each Uncut Pipe:

Waste for First 34 Pipes after Cutting 1 Struts Each: "

Number of Blue "A" Struts Created from the first 34 Pipes:

Number of Blue "A" Struts Created from the Last Pipe:

Waste on the Last Pipe after Cutting 1 Strut(s):

Total Waste in Creating 35 Blue "A" Struts:

Red "B" Struts

Red "B" Strut Units Entered: Feet and Inches: "

Number of Pipes needed to Create (30) Red "B" Struts:

Number of Red "B" Struts Produced From Each Uncut Pipe:

Waste for First 29 Pipes after Cutting 1 Struts Each: "

Number of Red "B" Struts Created from the first 29 Pipes:

Number of Red "B" Struts Created from the Last Pipe:

Waste on the Last Pipe after Cutting 1 Strut(s):

Total Waste in Creating 30 Red "B" Struts:

COST OF MATERIALS GRAND TOTALS:

Total Waste for All Struts:

Number of Uncut Pipe/Materials Needed:

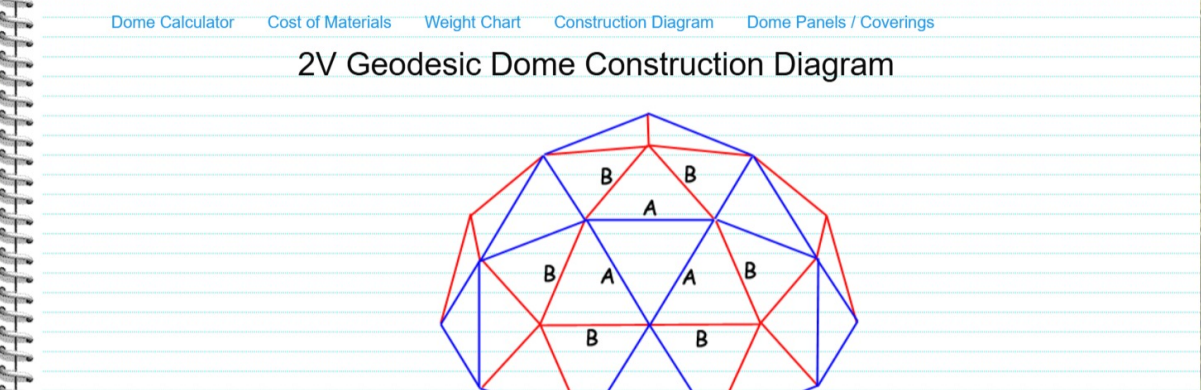
Total Cost of Materials @ \$9.50 Each:

[Instructional Video on How to Use This Section of the Dome Calculator](#)

Dome Weight Chart

Weight Chart	3/4" PVC	1" PVC	1.25" PVC	1.5" PVC	3/4" EMT	1" EMT	1.25" EMT	1.5" EMT	1" Bamboo	1.5" Bamboo
Lbs Per Linear Ft	.21	.32	.43	.51	.43	.64	.95	1.10	.2	.4
35 Blue "A" Struts	44.1	67.2	90.3	107.1	90.3	134.4	199.5	231	42	84
30 Red "B" Struts	33.4	50.9	68.4	81.2	68.4	101.9	151.2	175.1	31.8	63.7
Total Weight of Struts (lbs)	77.5	118.1	158.7	188.3	158.7	236.3	350.7	406.1	73.8	147.7
Weight of Hubs, Rings, Zip Ties	14.8	14.8	36.6	36.6	14.8	14.8	36.6	36.6	14.8	36.6
Total Dome Weight (lbs)	92.3	132.9	195.3	224.9	173.5	251.1	387.3	442.7	88.6	184.3

[Instructional Video on How to Use This Section of the Dome Calculator](#)



2 Frequency Geodesic Dome Tessellation Diagram

Blue "A" Chord Factor:

Red "B" Chord Factor:

Dome Panels / Coverings

A 2V Geodesic Dome Will Require Panels / Coverings for:

10 A-A-A Equilateral Triangles

30 B-A-B Isosceles Triangles

Total Triangles: 40

Triangle Dimensions

10 A-A-A Equilateral Triangles

A-A-A Triangle
Base = "A"

Altitude A:

Feet and Inches:

A-A-A Triangle Edge	Measurement	Strut Length	Center Ring Hub Width	Triangle Edge Total Length
A	Decimal	6'	0'	6'
	Feet and Inches	6' 0"	0"	6' 0"
Surface Area for Each A-A-A Triangle				15.588 Square Feet

30 B-A-B Isosceles Triangles

B-A-B Triangle
Base = "A"

Altitude A:

Feet and Inches:

B-A-B Triangle
Base = "B"

Altitude B:

Feet and Inches:

B-A-B Triangle Edge	Measurement	Strut Length	Center Ring Hub Width	Triangle Edge Total Length
A	Decimal	6'	0'	6'
	Feet and Inches	6' 0"	0"	6' 0"
B	Decimal	5.30586'	0'	5.30586'
	Feet and Inches	5' 3 - 5/8"	0"	5' 3 - 5/8"
Surface Area for Each B-A-B Triangle				13.129 Square Feet

[Instructional Video on How to Use This Section of the Dome Calculator](#)

2 Frequency Geodesic Dome Calculator generated by PERL on Monday, December 9, 2024.

Powered by [Notepad++](#).
Software Design by [johnhurt.com](#).

Responsive Mobile Design

This Site Uses

ZURB Foundation

The most advanced responsive front-end framework in the world.

Alba gu bràth - Scotland Until Judgment

To A Louse

"O, wad some Power the giffie gie us
To see oursel as others see us!
It wad frae monie a blunder free us,
An' foolish notion.
What airs in dress an' gait wad lea'e us,
An' ev'n devotion!"

Robert Burns - 1786

To place an order online, visit our [Online Catalog](#).

For a telephone order, call us at **1-615-390-3107**.

Or, go to our [Contact Us](#) page to send us an email.

Follow Zip Tie Domes on [FaceBook](#), [YouTube](#), [Pinterest](#) and [Instagram](#)

Contact us at: Zip Tie Domes
1857 Brindley Hollow Rd
Buffalo Valley, TN 38548 USA
Telephone: (615) 390-3107

Returns and Exchanges

Home - Geodesic Dome Plans - Geodesic Dome Kits - Geodesic "Hubs Only" Kits - Geodesic Chicken Coop Kits - Chicken Tractors For Sale - Geodesic Dome Greenhouse Kits - Geodesic Shelter Domes - Geodesic Silo Domes - Geodesic Pool Enclosure Dome Kits - Geodesic Tunnel Domes - Geodesic Octahedron Domes - Geodesic Half Domes - Geodesic Concrete Domes - Geodesic Spheres - Geodesic Dome Calculators - Geodesic Dome Construction Plans - Geodesic Sphere Calculators - Geodesic Dome Videos - Online Catalog - Frequently Asked Questions - Customer Reviews - Contact Us - Site Map - Geodesic "Connectors Only" Kits

Zip Tie Domes is a trademark of Zip Domes LLC.
© 2026 Zip Domes LLC All Rights Reserved.
This site is published by Zip Domes LLC which is solely responsible for its content.